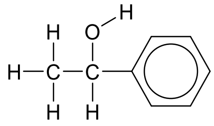
Structural formula

Draw the various structures of 1-phenylethanol

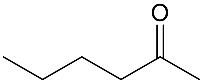
Condensed structural formula	Displayed formula/ Full structural formula	Skeletal formula	Stereochemical formula
$CH_3CH(OH)C_6H_5$			

Isomerism

Constitutional isomerism

- Definition:

  Constitutional isomers refer to molecules with the same molecular formula, but different ________________ formula



- Draw possible constitutional isomers of hexa-2-one, $C_6H_{12}O$

Chain isomerism

Different branching

Positional isomerism

Functional group at different position

Functional group isomerism

Different functional group

Stereoisomerism

Formula for total number of stereoisomers: 2^n; n = ____________________________

+ ____________________________

Enantiomerism

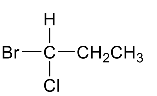
Reason / conditions: 1. no ________________________________

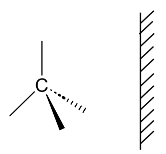
2. non- ________________________________

Differences in physical properties: Optical activity; enantiomers will ______________ plane polarised light in opposite directions

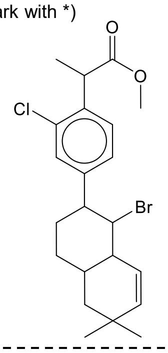
Enantiomers also differ in biological properties & in interaction with other chiral molecules

Draw the pair of enantiomers of the following compound





Identify all the chiral carbons (mark with *)



Cis-trans isomerism

Reason/ conditions: 1. ______________________ about a bond due to C=C or ring structure

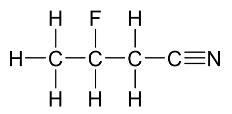
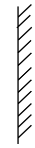
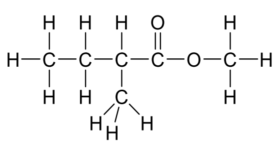
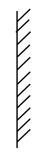
2. Each of the two C atoms (in C=C / in ring structure) are bonded to ______________________________

Differences in physical properties: melting point (________ higher)

boiling point (________ higher)

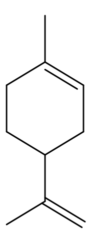
Draw and label the cis-trans isomers of $CH_3CH=CHCH_3$

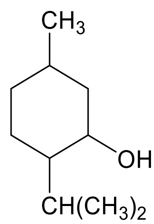
Extra Practice: Structural formula

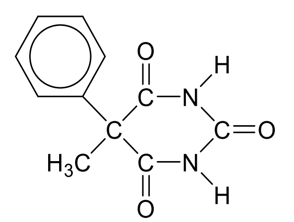
Condensed structural formula	Displayed formula/ Full structural formula	Skeletal formula	Stereochemical formula
			
$CH_3CH_2CHCHCH_2OH$			
			

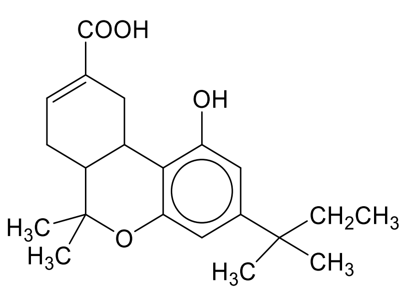
Extra Practice: Stereoisomerism

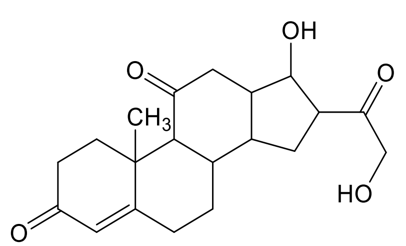
- Mark all chiral carbons with an (*)
- Calculate the number of stereoisomers that each of the compound have

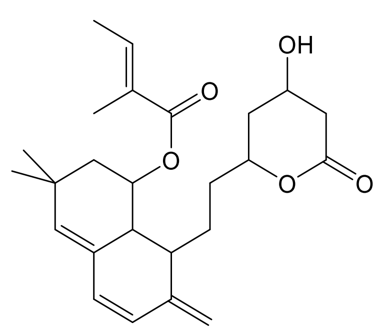






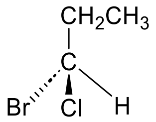


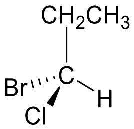


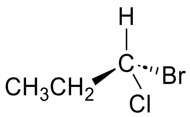


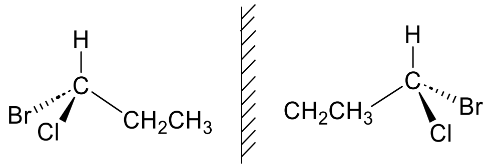
Extra Practice: Enantiomerism

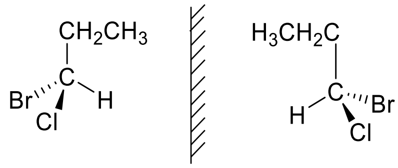
➢ The following sets of structures are drawn **incorrectly**. Identify the errors and draw the correct structures.





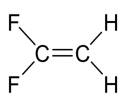
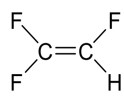
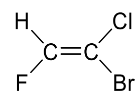
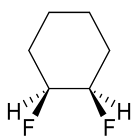
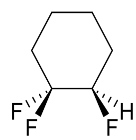
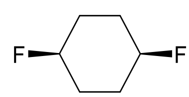
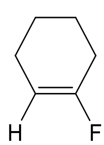
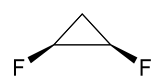
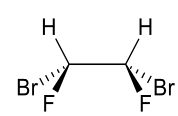






Extra Practice: Cis-trans isomerism

➢ Identify which of the following molecules exhibit cis-trans isomerism.

1 	2 	3 
4 	5 	6 
7  X	8 	9 

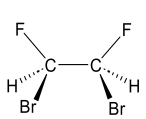
➢ Explain why (7) does not exhibit cis-trans isomerism

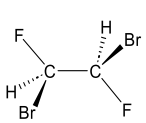
(7) is a cyclic alkene and its trans isomer would experience much ___________________.

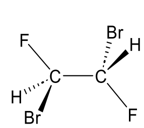
This makes it highly __________________ and therefore it does not exist.

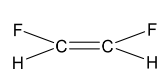
Extra Practice: Visualising 3D

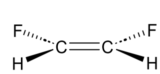
➢ Circle the identical species

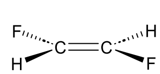


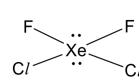


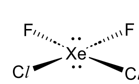


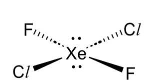






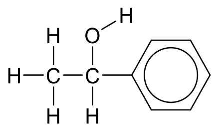
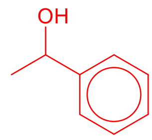
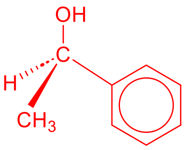






Structural formula

Draw the various structures of 1-phenylethanol

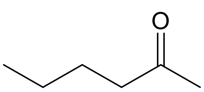
Condensed structural formula	Displayed formula/ Full structural formula	Skeletal formula	Stereochemical formula
$CH_3CH(OH)C_6H_5$			

Isomerism

Constitutional isomerism

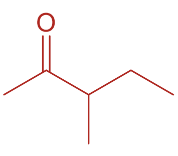
- Definition:

  Constitutional isomers refer to molecules with the same molecular formula, but different structural formula



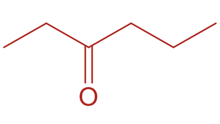
- Draw possible constitutional isomers of hexa-2-one, $C_6H_{12}O$

Chain isomerism



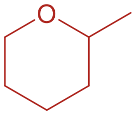
Different branching

Positional isomerism



Functional group at different position

Functional group isomerism



Different functional group

Stereoisomerism

Formula for total number of stereoisomers: 2^n n = no. of chiral carbon + no. of C=C that displays cis-trans

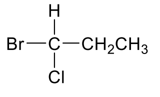
Enantiomerism

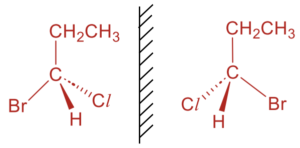
Reason / conditions:
1. no plane of symmetry
2. non-superimposable mirror image

Differences in physical properties: Optical activity; enantiomers will rotate plane polarised light in opposite directions

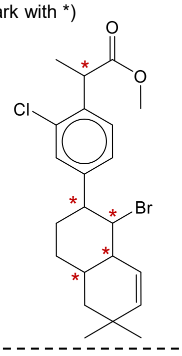
Enantiomers also differ in biological properties & in interaction with other chiral molecules

Draw the pair of enantiomers of the following compound





Identify all the chiral carbons (mark with *)

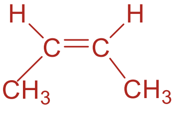


Cis-trans isomerism

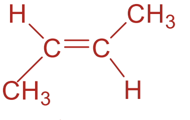
Reason/ conditions:
1. Restricted rotation about a bond due to π bond in C=C or ring structure
2. Each of the two C atoms (in C=C / in ring structure) are bonded to two different groups

Differences in physical properties: melting point (trans higher)
boiling point (cis higher)

Draw and label the cis-trans isomers of $CH_3CH=CHCH_3$

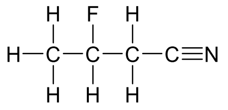
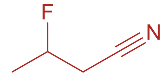
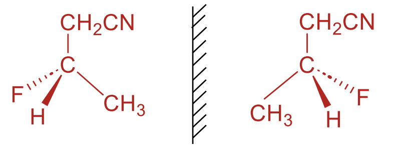
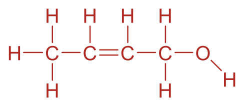
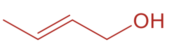
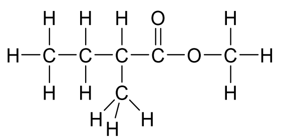
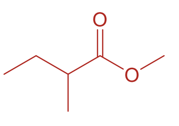
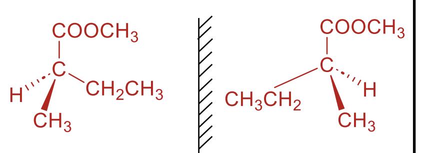


cis



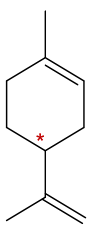
trans

Extra Practice: Structural formula

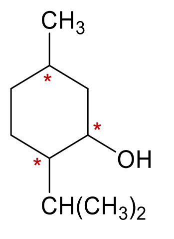
Condensed structural formula	Displayed formula/ Full structural formula	Skeletal formula	Stereochemical formula
CH_3CHFCH_2CN			 • complicated group at the top
$CH_3CHCHCH_2OH$	 • show O-H bond		-
$CH_3CH_2CH(CH_3)COOCH_3$			 • form bonds with the correct atoms

Extra Practice: Stereoisomerism

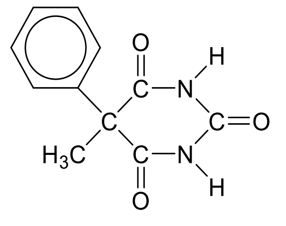
- Mark all chiral carbons with an (*)
- Calculate the number of stereoisomers that each of the compound have

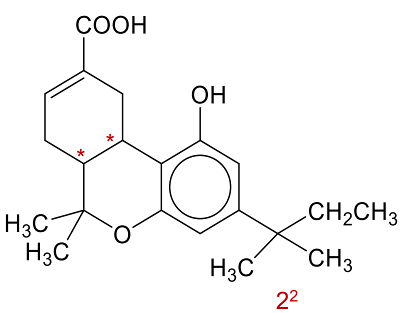


2

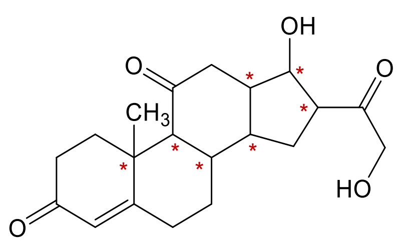


2^3

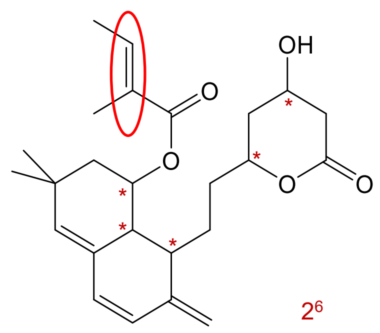




2^2



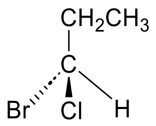
2^7



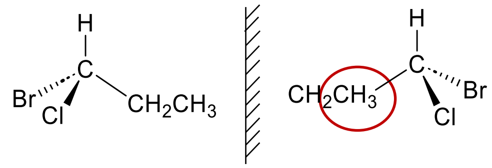
2^6

Extra Practice: Enantiomerism

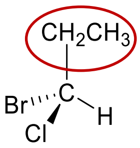
- The following sets of structures are drawn **incorrectly**. Identify the errors and draw the correct structures.



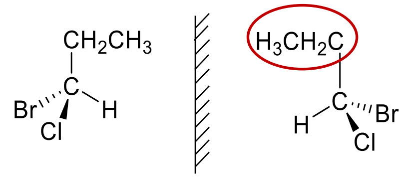
draw tetrahedral



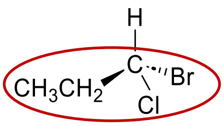
pointing to wrong C



point bond to C instead of H



not acceptable, difficult to interpret

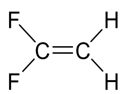
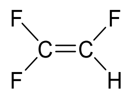
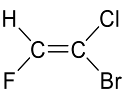
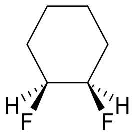
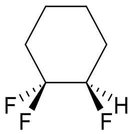
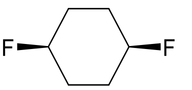
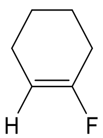
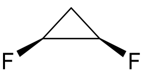
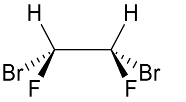


wedge and hash must be on same side

Tip: always draw the complicated structures on the top bond

Extra Practice: Cis-trans isomerism

- Identify which of the following molecules exhibit cis-trans isomerism.

(1)  X	(2)  X	(3)  ✓
(4)  ✓	(5)  X	(6)  ✓
(7)  X	(8)  ✓	(9)  X

- Explain why (7) does not exhibit cis-trans isomerism

(7) is a cyclic alkene and its trans isomer would experience much ring strain.

This makes it highly unstable and therefore it does not exist.

Extra Practice: Visualising 3D

- Circle the identical species